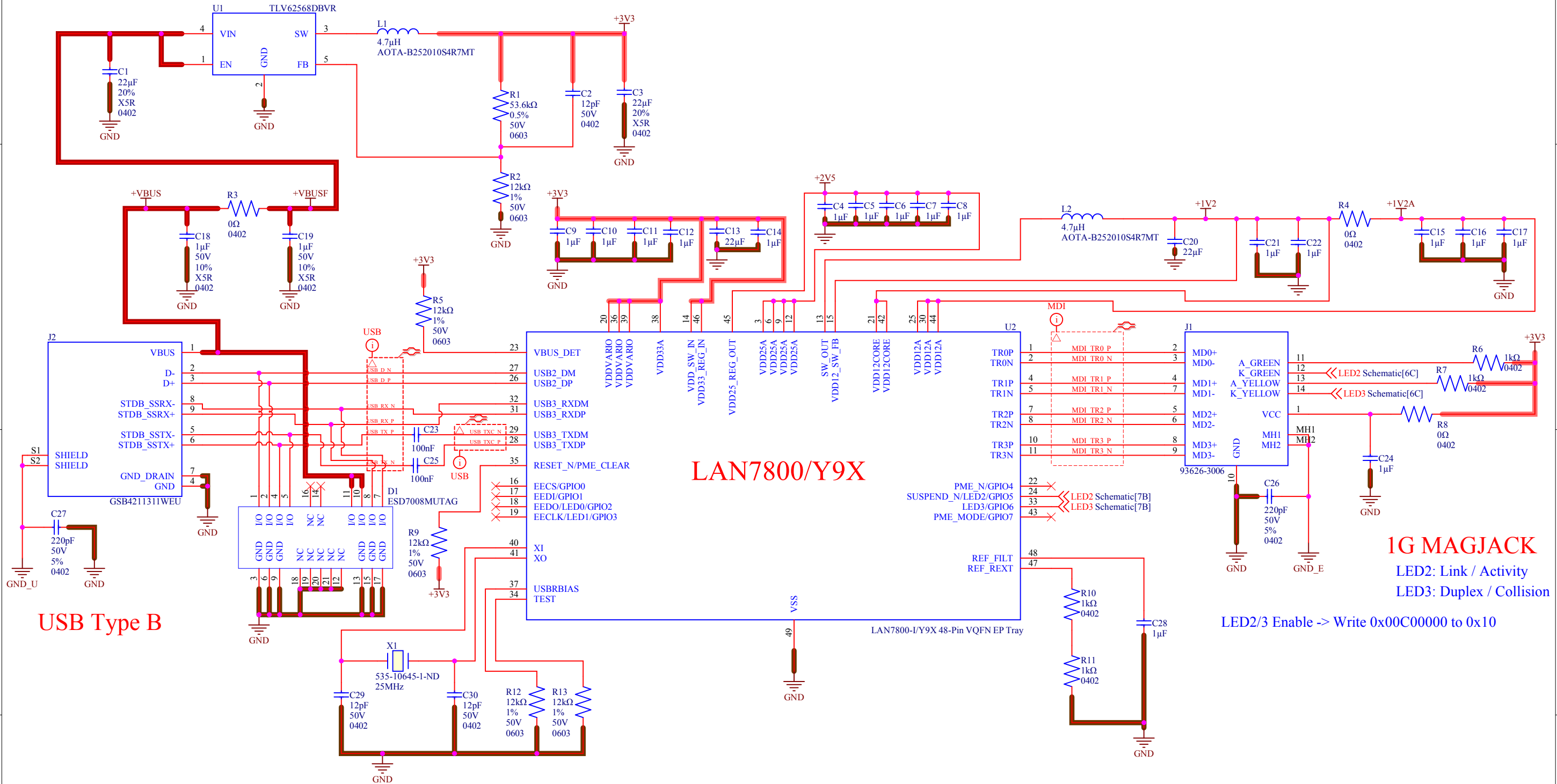


3V3 Buck Reguator

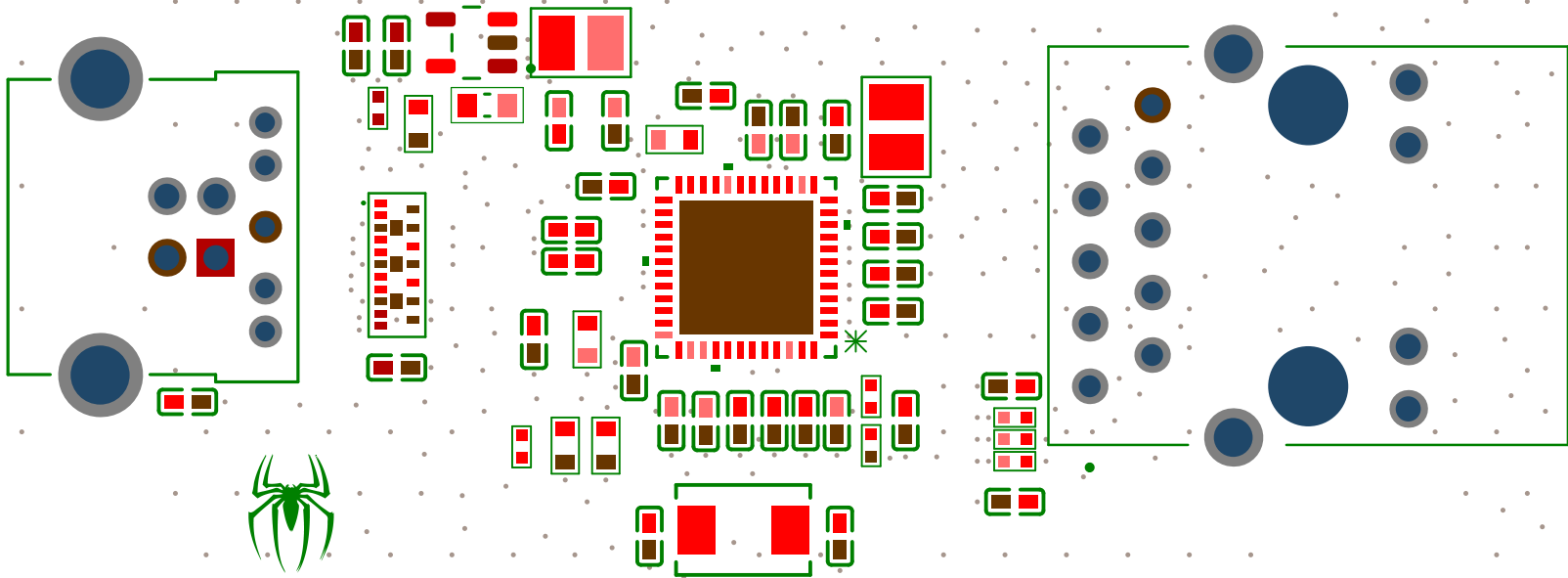
$$V_{out} = 0.6 * (1 + 53.6/12) = 3.28(V)$$



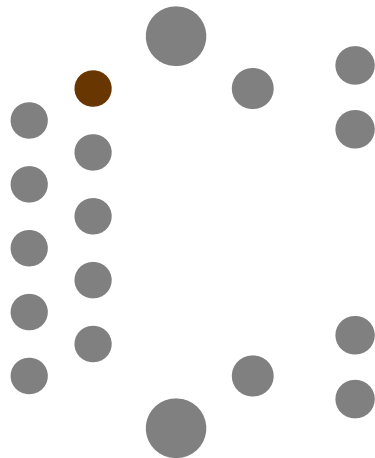
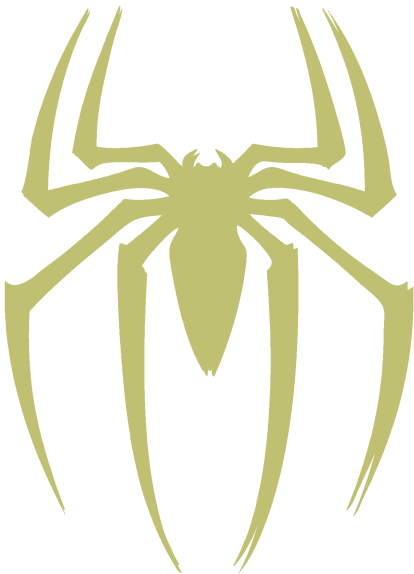
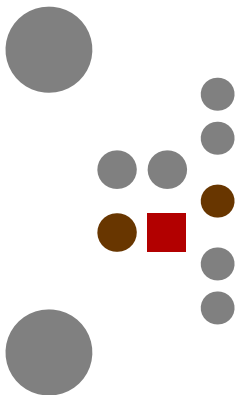
LED2/3 Enable -> Write 0x00C00000 to 0x10

Title: SCHEMATIC			Author: nq_minh_		Cannot open file D:\Admin\Github b\IoT_SOLUTIONS_LOAPAY\Documents\Template\LOGO_
Size: A3	Project: Adapter Hardware Design V1.0		CheckedBy:		
			Modified Date: 23/06/2026		
			Revision: V1.0		
Date: 23/06/2026	1:37:06 CH	Sheet 1 of 1		SW Version: 24.6.1.21	

USB 3.1 to Gigabit Ethernet



(c) Design by nq_minh_



Comment	Description	Designator	Footprint	Quantity	Manufacturer Part Number	LCSC Part	Manufacturer
22µF	6.3V 22µF X5R ±20% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C1, C3, C13, C20	C0402		4 GRM155R60J226M E11D	C415703	Murata
12pF	50V 12pF C0G ±1% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C2, C29, C30	C0402		3 GCM1555C1H120F A16D	C393682	Murata
1µF	Ceramic Capacitor, Multilayer, Ceramic, 50V, 10% +Tol, 10% - Tol, X5R, 1µF, Surface Mount, 0402	C4, C5, C6, C7, C8, C9, C10, C11, C12, C14, C15, C16, C17, C18, C19, C21, C22, C24, C28	C0402		19 GRM155R61H105K E05D	C1518208	Murata
100nF	50V 100nF X7R ±10% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C23, C25	C0402		2 CC0402KRX7R9BB 104	C131394	Yageo Group
220pF	50V 220pF C0G ±5% 0402 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C26, C27	C0402		2 GRM1555C1H221J A01D	C71693	Murata
ESD7008MUTAG	ESD Suppressor Diode TVS Bi-Dir 5V 10Vc 18-Pin UDFN EP T/R	D1	ESD7008MUTAG		1 ESD7008MUTAG	C260424	onsemi
93626-3006	Modular Connectors / Ethernet Connectors MXMag GIG BCORE RJ45 W/ LEDs GRN/YLW RA	J1	936263006		1 93626-3006	C6377799	Molex
GSB4211311WEU	USB Connector, 4 Contact(s), Female, Right Angle, Solder Terminal, Receptacle	J2	AMPHENOL_GSB4211 311WEU		1 GSB4211311WEU	C5429210	Amphenol
4.7µH	Inductor Power Mini Molded/Shielded Wirewound 4.7uH 20% 1MHz 1.6A 0.136Ohm DCR 1008	L1, L2	INDC2520X100N		2 AOTA- B252010S4R7MT	C24991964	Abracon
53.6kΩ	Fixed Resistor, Thin Film, 0.1W, 53600ohm, 75V, 0.5% +/-Tol, 50ppm/Cel, Surface Mount, 0603	R1	RESC1608X55N		1 RT0603DRE0753K6 L	C863153	Yageo Group
12kΩ	100mW Thick Film Resistors 75V ±100ppm/°C ±1% 12kΩ 0603 Chip Resistor - Surface Mount ROHS	R2, R5, R9, R12, R13	RC0603N_YAG-L		5 RC0603FR-0712KL	C114659	Yageo Group
0Ω	GENERAL PURPOSE CHIP RESISTOR Metal Glaze/thick Film, 0.0625W, 0ohm, Surface Mount, 0402	R3, R4, R8	RC0402N_YAG-L		3 RC0402JR-070RL	C60485	Yageo Group
1kΩ	RES SMD 1K OHM 0.1% 1/16W 0402 / 1 kOhms ±0.1% 0.063W, 1/16W Chip Resistor 0402 (1005 Metric) Thin Film	R6, R7, R10, R11	RC0402N_YAG-L		4 RT0402BRD071KL	C852624	Yageo Group
TLV62568DBVR	ROHS3Compliant Adjustable Positive DUAL Switching Regulator IC - 40C~125C TJ 1.5MHz 1A 1.6mm	U1	SOT95P280X145-5N		1 TLV62568DBVR	C163219	Texas Instruments
LAN7800-I/Y9X 48-Pin VQFN EP Tray	Ethernet CTLR Single Chip 10Mbps/100Mbps/10 00Mbps 1.8V/3.3V Automotive 48-Pin VQFN EP Tray	U2	SQFN48_7X7_Y9X_MC H-L		1 LAN7800-I/Y9X	C633489	Microchip
ABM3-25.000MHZ- D2Y-T 25Mhz	Crystal 25MHz ±20ppm (Tol) ±30ppm (Stability) 18pF FUND 300hm 2-Pin Mini- CSMD T/R	X1	XTAL_ABM3_ABR-L		1 ABM3-25.000MHZ- D2Y-T	C148312	Abracon